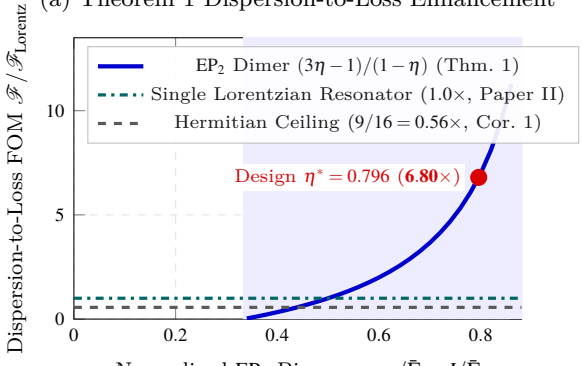
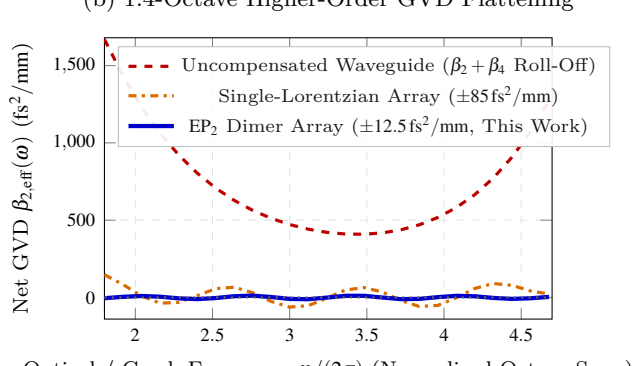

(a) Theorem 1 Dispersion-to-Loss Enhancement
EP$_2$ Dimer $(3\eta-1)/(1-\eta)$ (Thm. 1)
Single Lorentzian Resonator (1.0×, Paper II)
Hermitian Ceiling $(9/16 = 0.56\times$, Cor. 1)
Design $\eta^* = 0.796$ (**6.80**×)
Dispersion-to-Loss FOM $\mathscr{F}/\mathscr{F}_{\text{Lorentz}}$
Normalized EP$_2$ Bias $\eta = \gamma_{\text{PT}}/\bar{\Gamma} = J/\bar{\Gamma}$
0
5
10
0
0.2
0.4
0.6
0.8


(b) 1.4-Octave Higher-Order GVD Flattening
Uncompensated Waveguide ($\beta_2+\beta_4$ Roll-Off)
Single-Lorentzian Array (±85 fs$^2$/mm)
EP$_2$ Dimer Array (±12.5 fs$^2$/mm, This Work)
Net GVD $\beta_{2,\text{eff}}(\omega)$ (fs$^2$/mm)
Optical / Comb Frequency $\omega/(2\pi)$ (Normalized Octave Span)
1,500
1,000
500
0
2
2.5
3
3.5
4
4.5